



PHYSICS SSC-I

Inclusive Scheme of Studies 2024 (1200 Marks)

SECTION - A (Marks 12)

Time allowed: 15 Minutes

Candidate Sign. _____

Invigilator Sign. _____

Answer Sheet No. _____

Version No.				
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Deleting/overwriting is not allowed.
Do not use lead pencil.

$$3 \times 10^3 \times 10^{-9} = 3 \times 10^{-6}$$

$$V = \frac{2 \times 10^9}{4 \times 60}$$

Fill the relevant bubble against each question according to curriculum:

Question	A	B	C	D	A	B	C	D
1. A device measures a charge of 3000 nC. Convert it into microcoulombs (μC).	0.03 μC	3 μC	30 μC	300 μC	☐	☐	☐	☐
2. A cyclist moves 2 km in 4 minutes. What is the speed in m/s?	4 m/s	8.3 m/s	12 m/s	16.3 m/s	☐	☐	☐	☐
3. Which fundamental force is the strongest of all?	Gravitational	Electro-weak	Strong force	Magnetic	☐	☐	☐	☐
4. What is a reasonable estimate for the mass of a physics textbook?	2 mg	1 kg	10 kg	1 g	☐	☐	☐	☐
5. Which unit belongs to a derived quantity?	Second (s)	Kelvin (K)	Meter (m)	Newton (N)	☐	☐	☐	☐
6. Two unequal forces act parallel but in opposite directions, the forces are:	Unlike parallel forces	Balanced forces	Like parallel forces	Centripetal forces	☐	☐	☐	☐
7. A hypothesis is best described as:	A testable prediction	A universally accepted fact	A mathematical equation	A proven theory	☐	☐	☐	☐
8. When a soft iron piece is placed near a magnet, it:	Becomes a permanent magnet	Loses its mass	Gets temporary poles (N and S)	Repels the magnet	☐	☐	☐	☐
9. Why does a heavy object resist change in its motion?	It has greater inertia	It has more speed	It has more friction	It has no force	☐	☐	☐	☐
10. If a man applies 50 N of force to lift a load 0.5 m upward then what is the work done?	10 J	20 J	25 J	50 J	☐	☐	☐	☐
11. Which one of the following is the correct formula for magnetic field of a solenoid?	$B = \frac{\mu n}{\ell}$	$B = \mu_0 n \ell$	$B = \frac{n \ell}{\mu_0}$	$B = n_0 \ell$	☐	☐	☐	☐
12. A scientist who studies matter, energy, and their interactions is called a:	Chemist	Biologist	Engineer	Physicist	☐	☐	☐	☐

1SA-I 26004-(B) —

• $v = \frac{d}{t}$ • $W = \vec{F} \cdot \vec{d}$ • $w = mg$



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Fill the relevant bubble against each question according to curriculum:

Question	A	B	C	D	A	B	C	D
1. Which of the following is a derived quantity?	Mass	Speed	Length	Temperature	☐	☐	☐	☐
2. The SI unit of energy is:	Joule	Pascal	Watt	Newton	☐	☐	☐	☐
3. The movement of a pendulum is an example of:	Linear motion	Rotatory motion	Vibratory motion	Circular motion	☐	☐	☐	☐
4. According to Newton's 3 rd law, when someone pushes a concrete wall, the wall:	Does nothing	Moves backward	Pushes back	Pulls forward	☐	☐	☐	☐
5. The branch of physics that studies atomic nuclei and their reactions is known as:	Nuclear Physics	Quantum Mechanics	Thermal Physics	Biophysics	☐	☐	☐	☐
6. A spring extends 3 cm when a 6 N force is applied. What is its spring constant?	18 N/m	200 N/m	50 N/m	2 N/m	☐	☐	☐	☐
7. Which statement best describes a magnetic field?	It is visible in air	It acts only at long distances	It only exists inside metals	It surrounds a magnet	☐	☐	☐	☐
8. A hypothesis is best described as:	A proven law	A testable prediction	A mathematical rule	A universal fact	☐	☐	☐	☐
9. Why does a heavy object resist change in its motion more than a light object?	It has greater inertia	It moves faster	It has more friction	It has more energy	☐	☐	☐	☐
10. A body of mass 4 kg has a kinetic energy of 200 J. What is its speed?	5 m/s	10 m/s	20 m/s	40 m/s	☐	☐	☐	☐
11. Which formula correctly relates force and momentum?	$F = mvt$	$F = \frac{m}{vt}$	$F = \frac{mt}{v}$	$F = \frac{mv}{t}$	☐	☐	☐	☐
12. A temporary magnet is usually made of:	Rubber	Copper	Soft iron	Plastic	☐	☐	☐	☐

—1SA-I 26004-(D)—



PHYSICS SSC-I

Inclusive Scheme of Studies 2024 (1200 Marks)

SECTION – A (Marks 12)

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Candidate Sign. _____

Invigilator Sign. _____

Answer Sheet No. _____

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Fill the relevant bubble against each question according to curriculum:

Question	A	B	C	D	A	B	C	D
1. Which pair correctly shows base quantity and its unit?	Temperature – kg	Time – m	Mass – kg	Speed – s	☐	☐	☐	☐
2. When a piece of soft iron is placed near a magnet, it becomes:	Permanently magnetized	Temporarily magnetized	Demagnetized	Electrically charged	☐	☐	☐	☐
3. The study of electricity and magnetism together is known as:	Electromagnetism	Nuclear Physics	Classical Mechanics	Particle Physics	☐	☐	☐	☐
4. A car travels 150 m in 10 seconds. What is its speed?	5 m/s	10 m/s	15 m/s	20 m/s	☒	☐	☐	☐
5. A 4 kg object reaches the ground with a kinetic energy of 156.8 J. Ignoring air resistance, from what height was it dropped? (Use $g = 9.8 \text{ m/s}^2$)	2 m	3 m	4 m	5 m	☐	☐	☐	☐
6. Which statement best explains why displacement is a vector quantity?	It has only magnitude	It depends on path length	It has magnitude and direction	It is measured in meters	☐	☐	☐	☒
7. When a north pole of a magnet is placed in an external magnetic field, it:	Always moves toward east	Experiences a force	Becomes a south pole	Stops producing a field	☐	☐	☐	☒
8. A gun recoils backward when a bullet is fired because:	The gun is heavier	Air pushes the gun	The bullet loses speed	Reaction forces act on gun	☐	☐	☐	☒
9. If a bar magnet is cut in half it will become:	The same magnet	Un-magnetized	A monopole	Magnet of less strength	☐	☐	☐	☐
10. In scientific research, a theory is:	A guess with no evidence	A personal opinion	A rule that never changes	Explanation based on experiments	☐	☐	☐	☒
11. A ball dropped freely from a tower reaches ground in 4s, the speed of impact of ball is:	0 m/s	2.45 m/s	19.6 m/s	39.2 m/s	☐	☐	☐	☒
12. A mosquito flying in different directions shows:	Linear motion	Random motion	Circular motion	Vibratory motion	☐	☒	☐	☐

1SA-I 26004-(M) –

$$v = \frac{d}{t}$$

$$PE = mgh$$

$$g = 9.8 \text{ m/s}^2$$

$$g = \frac{v_f - v_i}{\Delta t}$$



PHYSICS SSC-I

SECTION – A (Marks 12)

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حصہ اول لازمی ہے۔ اس کے جوابات اسی صفحہ پر دے کر ناظم مرکز کے حوالے کریں۔ کاٹ کر دوبارہ گنتے کی اجازت نہیں ہے۔ لید پینسل کا استعمال منع ہے۔

Version No.				
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Answer Sheet No. _____

Invigilator Sign. _____

Fill the relevant bubble against each question according to curriculum:

Candidate Sign. _____

Question	سوال	A	B	C	D	A	B	C	D
1. The SI unit of momentum is:	مومنٹم کی SI کا یونٹ کیا ہے؟	$kgms^{-2}$	kgm	Ns	Ns^{-1}	☐	☒	☐	☐
2. Which one is a scalar quantity?	درج شدہ میں سے کون سی سکالر مقدار ہے؟	velocity	force	energy	weight	☐	☐	☒	☐
3. In 5s a car accelerates so that its velocity increases by $20ms^{-1}$. the acceleration is:	5 سیکنڈ میں ایک کار اتنی تیز ہوتی ہے کہ اس کی دلائی 20 میٹر فی سیکنڈ بڑھ جاتی ہے اس کی ایکسلریشن کتنی ہوگی؟	$0.25ms^{-2}$	$4ms^{-2}$	$25ms^{-2}$	$100ms^{-2}$	☐	☐	☐	☒
4. The working of an air-conditioner is primarily related with:	ایئر کنڈیشنر کی درگت بنیادی طور پر _____ کا حصہ ہے۔	Electromagnetism	Thermodynamics	Kinematics	Dynamics	☐	☐	☐	☐
5. The rate of change of velocity is called:	دلائی میں تبدیلی کی شرح کو کیا کہتے ہیں؟	speed	acceleration	force	motion	☐	☐	☐	☐
6. According to Newton's second law of motion:	نیوٹن کے قانون حرکت کے مطابق:	$a \propto m$	$a \propto \frac{1}{F}$	$a \propto F$	$a \propto \frac{m}{F}$	☐	☐	☐	☐
7. A bar magnet's magnetic field lines emerge from:	بار میگنٹ کی میگنیٹک لائنز کہاں سے نکلتی ہیں؟	North pole	South pole	Center	Random	☐	☐	☐	☐
8. Which of the following is a non-contact force?	درج شدہ میں سے کون سی ٹان کا میٹک فورس ہے؟	Tension	Normal force	Friction	Magnetic force	☐	☐	☐	☒
9. The branch of physics that deals with the behavior of microscopic particles is:	فزکس کی کون سی شاخ مائیکروسکوپک پارٹیکلز کا مطالعہ کرتی ہے؟	Electricity	Magnetism	Optics	Quantum Mechanics	☐	☐	☐	☒
10. Which of the following material is ferromagnetic?	درج شدہ میں سے کون سا میٹیریل فیرو میگنیٹک ہے؟	Silver	Copper	Aluminum	Nickel	☐	☐	☐	☒
11. Work is done only when:	ورک صرف تب ہوتا ہے جب:	Force causes displacement	Force acts only	Only time is spent	Motion is zero	☐	☐	☐	☐
12. Energy possessed due to motion is called:	حرکت کی وجہ سے موجود توانائی کو کیا کہتے ہیں؟	Potential energy	Nuclear energy	Thermal energy	Kinetic energy	☐	☐	☐	☒



PHYSICS SSC-I

SECTION – A (Marks 12)

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گھسنے کی اجازت نہیں ہے۔ سپر پینل کا استعمال ممنوع ہے۔

Version No.				
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Answer Sheet No. _____

Invigilator Sign. _____

Fill the relevant bubble against each question according to curriculum:

Candidate Sign. _____

Question	سوال	A	B	C	D	A	B	C	D
1. 1 picogram =		$10^{-3} g$	$10^{-6} g$	$10^{-12} g$	$10^{-15} g$	☐	☐	☐	☐
2. A bird accelerates at $0.3ms^{-2}$ for 3s. Δv is :		$0.9ms^{-1}$	$1.5ms^{-1}$	$1.95ms^{-1}$	$2.4ms^{-1}$	☐	☐	☐	☐
3. The given graph shows:		Uniform velocity یکساں ولاسٹی	Uniform acceleration یکساں ایکسلریشن	Variable acceleration متغیر ایکسلریشن	Negative acceleration منفی ایکسلریشن	☐	☐	☐	☐
4. A force changes momentum of a body by $\Delta p = 50kgms^{-1}$ in 2 seconds then force $F =$		100 N	50 N	25 N	2 N	☐	☐	☐	☐
5. Torque in a rotating fan of blade length 0.5 m due to a 1N force is:		0.5 Nm	2 Nm	-0.5 Nm	0 Nm	☐	☐	☐	☐
6. $F = 15N$ and $x = 1.5m$ then spring constant 'k' is:		$10Nm^{-1}$	10Nm	$0.1Nm^{-2}$	$5N^{-1}m^{-1}$	☐	☐	☐	☐
7. A machine lifts a 200N load to a height of 30m in 50 seconds. Average power of machine is:		1.33 W	60 W	120 W	6000 W	☐	☐	☐	☐
8. Ability of thermometer to show temperature variations uniformly and equally is called:		Range	Linearity	Sensitivity	Reliability	☐	☐	☐	☐
9. An electromagnet can be found in:		Refrigerator	Digital clock	Solar panel	Electric fan	☐	☐	☐	☐
10. In loudspeakers the material used is:		Diamagnetic	Paramagnetic	Ferromagnetic	Non-magnetic	☐	☐	☐	☐
11. Studying how air conditioner cools a house, is a part of:		Thermodynamics	Electromagnetism	Nuclear physics	Optics	☐	☐	☐	☐
12. Which branch of physics deals with transmission and control of sound?		Astronomy	Cosmetology	Optics	Acoustics	☐	☐	☐	☐

$a = \frac{\Delta v}{\Delta t}$ $F = \frac{\Delta p}{\Delta t}$ $\tau = IF$ $P = \frac{mgh}{t}$ $F = kx$ $\Delta p = F\Delta t$ $Weight = mg$

$k = \frac{F}{x}$



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Answer Sheet No. _____

Invigilator Sign. _____

Fill the relevant bubble against each question according to curriculum:

Candidate Sign. _____

Question	سوال	A	B	C	D	A	B	C	D
1. 1 milligram =		$10^{-3} g$	$10^{-6} g$	$10^{-12} g$	$10^{-15} g$	☐	☐	☐	☐
2. A car changes its velocity from $20ms^{-1}$ to $56ms^{-1}$ in 2s. Its acceleration is:	ایک کار کی دلائی 2s میں $20ms^{-1}$ سے $56ms^{-1}$ ہو تو اس کی ایکسیریشن:	$28ms^{-2}$	$36ms^{-2}$	$10ms^{-2}$	$18ms^{-2}$	☐	☐	☐	☐
3. The given graph shows:	Speed v Time t	Zero acceleration صفر ایکسیریشن	Uniform acceleration یکساں ایکسیریشن	Variable acceleration متغیر ایکسیریشن	Negative acceleration منفی ایکسیریشن	☐	☐	☐	☐
4. A motorcycle of mass 80kg moving with speed $20ms^{-1}$ hits a wall with a force of 2000N. The time of collision is:	ایک 80kg ماس کا موٹر سائیکل $20ms^{-1}$ کی سپیڈ سے ایک دیوار سے 2000N کی قوت سے ٹکراتا ہے۔ ٹکراؤ کا وقت معلوم کریں۔	0.4 seconds 0.4 سیکنڈ	0.6 seconds 0.6 سیکنڈ	0.8 seconds 0.8 سیکنڈ	0.10 seconds 0.10 سیکنڈ	☐	☐	☐	☐
5. A spanner of length 30cm uses a force of 150N to open a tight nut. The torque is:	ایک ریچ جس کی لمبائی 30 سینٹی میٹر ہے اس سے ایک نٹ کھولنے کے لیے 150 N کی قوت لگائی جاتی ہے تو ٹکڑا ٹاک پیدا ہوگا؟	10 Nm	25 Nm	35 Nm	45 Nm	☐	☐	☐	☐
6. If $F=25N$ and $x=2.5m$ then spring constant 'k' is:	اگر $F=25N$ اور $x=2.5m$ ہوں تو سپرنگ کانٹنٹ 'k' کی کیا بلیو ہوگی؟	$0.1Nm^{-1}$	10Nm	$5Nm^{-2}$	$10Nm^{-1}$	☐	☐	☐	☐
7. In how much time will a 120W motor lift 200N weight vertically up to a height of 30m?	120 واٹ کی موٹر 200N وزن کو 30 میٹر تک کھینچنے کے وقت میں اٹھائے گی؟	50 seconds 50 سیکنڈ	60 seconds 60 سیکنڈ	70 seconds 70 سیکنڈ	80 seconds 80 سیکنڈ	☐	☐	☐	☐
8. Ability of thermometer to show temperature variations uniformly and equally is called:	تھرمو میٹر کی ٹیمپریچر میں تبدیلی کو یونیفارم اور یکساں دکھانے کی صلاحیت کیا کہلاتی ہے؟	Range رینج	Linearity لینیریٹی	Sensitivity سینسٹیویٹی	Reliability ریلیائیبلٹی	☐	☐	☐	☐
9. Electromagnet is NOT found in:	ورن شدہ میں سے کس میں الیکٹرو میگنیٹ نہیں پایا جاتا؟	A digital clock ایک ڈیجیٹل کلاک	Electric motor الیکٹرک موٹر	Loudspeaker لاؤڈ سپیکر	Electric fan برقی کھانکھا	☐	☐	☐	☐
10. Magnetism field is strongest where magnets filed lines are:	مگنیٹک فیلڈ اس مقام پر زیادہ ستر وگ ہوتی ہے جہاں مگنیٹک فیلڈ لائنز:	Closest بہت قریب ترین	Farthest بہت دور ترین	Mutually perpendicular ایک دوسرے پر عمود	Parallel ایک دوسرے کے ساتھ	☐	☐	☐	☐
11. The working of a deep freezer is primarily related with:	ڈیپ فریڈر کی ورکنگ بنیادی طور پر _____ کا حصہ ہے۔	Electromagnetism الیکٹرو میگنیٹزم	Thermodynamics تھرمو ڈائنامکس	Kinematics کائنا میٹکس	Dynamics ڈائنامکس	☐	☐	☐	☐
12. Which branch of physics deals with nature and properties of light?	فزکس کی کس برانچ میں لائٹ کی فطرت اور خصوصیات سٹڈی کرتے ہیں؟	Astronomy آسٹرونومی	Cosmetology کاسمیٹالوجی	Acoustics اکاؤسٹکس	Optics آپٹکس	☐	☐	☐	☐

$v_f = v_i + at$ $P = \frac{mgh}{t}$ $\tau = Fl$ $F = ma$ $P = \frac{F}{A}$ $F = kx$ $F = \frac{\Delta p}{\Delta t}$



Federal Board SSC-I Examination

Physics Model Question Paper

Curriculum 2022-2023 (Inclusive Scheme of Studies 2024)

Section - A (Marks 12)

Time Allowed: 20 minutes

Section – A is compulsory. All parts of this section are to be answered on this page and handed over to the Centre Superintendent. Deleting/overwriting is not allowed. Do not use lead pencil.

ROLL NUMBER					
0	0	0	0	0	0
1	1	1	1	1	1
2	2	2	2	2	2
3	3	3	3	3	3
4	4	4	4	4	4
5	5	5	5	5	5
6	6	6	6	6	6
7	7	7	7	7	7
8	8	8	8	8	8
9	9	9	9	9	9

Version No.			
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Candidate Sign. _____

Invigilator Sign. _____

Q1. Fill the relevant bubble against each question according to curriculum. Each part carries one mark.

S #	Question	(A)	(B)	(C)	(D)	(A)	(B)	(C)	(D)
1.	When a body moves in a circular path, its velocity is:	Constant	Variable	Zero	Increasing	☐	☐	☐	☐
2.	The force which opposes the relative motion between two surfaces in contact is known as:	Friction	Gravitational force	Electrostatic force	Nuclear force	☐	☐	☐	☐
3.	A train is traveling at 20 m/s and comes to a stop in 40 seconds. What is the magnitude of deceleration of the train?	0.5 m/s ²	2 m/s ²	0.05 m/s ²	0.2 m/s ²	☐	☐	☐	☐
4.	If two objects, one heavy and one light, are dropped from the same height, neglecting air resistance, which object will hit the ground first?	The heavy object	The light object	Both objects will hit the ground at the same time	It depends on the shape of the objects	☐	☐	☐	☐
5.	What will be weight of a 5kg object on moon where gravitational acceleration is 1.6 ms ⁻² ?	1.6 N	8 N	9.8 N	80 N	☐	☐	☐	☐
6.	When electric current passes through an air-cored coil, the compass needle placed nearby deflects due to the magnetic field. If an iron nail is inserted into the coil, the compass needle will:	Not deflect at all	Deflect less and slowly	Deflect more quickly and strongly	Point only in the direction of Earth's magnetic field	☐	☐	☐	☐
7.	Which of the followings is a greatest prefix?	Deca 10 ¹	Deci 10 ⁻¹	Milli 10 ⁻³	Nano 10 ⁻⁹	☐	☐	☐	☐
8.	joule (J) is the unit of work which is equal to:	Newton	kg m s ⁻²	Watt second	Newton second	☐	☐	☐	☐
9.	A car, an elephant and a cricket ball have same kinetic energies. Which of these have greater speed?	Car	Elephant	Cricket	All have same speed	☐	☐	☐	☐
10.	Magnetic field lines:	Are farthest at poles	Intersect each other	Never intersect each other	Do not pass in vacuum	☐	☐	☐	☐
11.	The Branch of Physics that is most important when studying how glasses help people see:	Thermodynamics	Electromagnetism	Mechanics	Optics	☐	☐	☐	☐
12.	Which term describes a thoroughly tested idea in physics?	Idea	Hypothesis	Theory	Law	☐	☐	☐	☐



Federal Board SSC-I Examination
Physics Model Question Paper
National Curriculum 2022-2023 (Inclusive Scheme of Studies 2024)
(50% MCQs & 50% Descriptive)

SECTION - A (Marks 30)

Time Allowed: 1:20 hours

Section – A is compulsory. All parts of this section are to be answered on the OMR Sheet given in the E-Sheet and handed over to the Centre Superintendent.
Deleting/ overwriting is not allowed. Do not use lead pencil.

Roll Number						Paper Version
						A

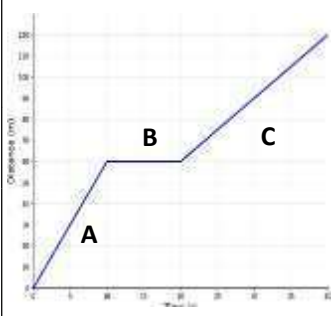
Candidate Sign. _____

Invigilator Sign. _____

Note: Fill the correct bubble for each question on OMR Sheet given in the E-Sheet. Each question carries 0.5 mark.

S #	Question	(A)	(B)	(C)	(D)
1.	Which of the following is a base quantity?	Velocity	Mass	Force	Momentum
2.	The rate of change of displacement is called:	Speed	Velocity	Acceleration	Distance
3.	Which of the following forces is a non-contact force?	Friction	Tension	Magnetic force	Air resistance
4.	The tendency of an object to resist a change in its state of rest or motion is called:	Force	Weight	Inertia	Momentum
5.	Pascal's Law is applicable to:	Gases only	Solids only	Fluids	Vacuum
6.	Which of the following is a scalar quantity?	Force	Displacement	Energy	Velocity
7.	The SI unit of momentum is:	N m	kg m/s	kg m/s ²	J
8.	Which statement best describes translational motion of a body?	all its parts move in one direction	rotates about a fixed axis	moves about a fixed point	Changes its shape
9.	The value of 'g' at the surface of the moon is approximately:	9.8 m/s ²	1.6 m/s ²	Infinite	0 m/s ²
10.	Two identical beakers have same amount of water. Beaker A is at 30°C while beaker B is at 60°C temperature. This is because,	Particles of beaker B has more Internal energy	Particles of beaker A has more density	Particles of beaker A has more Internal energy	Particles of beaker B has less density

11.	A book is lifted from the floor to a shelf. It gains gravitational potential energy (P.E). Which factor is responsible for the gain in P.E of book?	Energy lost by the book as heat	Work done against gravity to raise the book	The increase in the speed of book	The increase in the weight of book
12.	Which one of the following is a renewable energy source?	Coal	Natural gas	Solar energy	Nuclear energy
13.	A group of students carries a liquid barometer while climbing a mountain. They observe that the mercury level decreases as they go higher. What is the reason for this decrease?	Lower air pressure	Stronger gravity	Lower temperature	Higher air pressure
14.	The center of gravity of a uniform rod lies at:	it's one end	its center	One-quarter from an end	outside the rod
15.	What is the purpose of using ball bearings in a machine?	they increase friction	they reduce friction	they increase mass	they reduce air resistance
16.	A file size is 2 gigabytes (GB). If expressed in megabytes (MB), what is the correct value?	2 MB	20 MB	200 MB	2000 MB
17.	Which factor determines validity of a theory?	Length of time	Experimental evidence	Popularity	Number of equations
18.	In a hydraulic press, the pressure applied on the small piston is:	decreased	transmitted equally	increased	vanished
19.	The energy stored in a compressed spring is:	Kinetic energy	Elastic potential energy	Chemical energy	Thermal energy
20.	The slope of a distance-time graph gives:	Acceleration	Force	Speed	Work
21.	A shopping cart is pushed with a force of 2 N for 5 s. What impulse is given to the cart?	2 Ns	5 Ns	10 Ns	20 Ns
22.	A factory machine has an efficiency of 30%. It means that:	30% of the input energy is lost as heat	30% of the input energy is converted into useful output	70% of the input energy is converted into useful output	40% of the input energy is converted into useful output
23.	When a pressure is applied on a surface, the force acts:	Parallel to the surface	Perpendicular to the surface	At 45° angle to the surface	In the same direction as gravity

24.	When a bus starts suddenly, passengers fall backward due to:	Friction	Weight	Inertia	Gravitation
25.	An astronaut takes an object that weighs 200 N on Earth to an unknown planet, where it weighs 50N. If the mass of the object is 20kg, what is the gravitational field strength on that planet?	0.4N/kg	9.8 N/kg	2.5 N/kg	25 N/kg
26.	Like parallel forces act in:	opposite directions	same direction	at an angle of 90° to each other	at an angle of 45° to each other
27.	The state of matter found in stars is called:	Solid	Plasma	Gas	Liquid
28.	How much work is needed to lift a bag of groceries that weighs 150 N to a height of 4 m?	300 J	450 J	600 J	750 J
29.	Vector X points towards east and vector Y points towards north. In which direction does the resultant vector lie?	North - East	South-East	North	East
30.	The branch of physics which deals with study of light is:	Mechanics	Optics	Thermodynamics	Magnetism
31.	In which of the following situations are the distance traveled and the magnitude of the displacement equal?	A child moving on a swing	An object is moving along a circular path.	A car moving on a curved road	Linear motion in one direction.
32.	When an external magnetic field is removed, a paramagnetic material usually:	Remains permanently magnetized	Loses most of its induced magnetism	Becomes strongly magnetic	Repels magnetic fields strongly
33.	The distance–time graph for a cyclist moving along a straight road is shown. Which motion is displayed by the cyclist in portion A of the graph? 	At rest	Uniform speed	Non-uniform speed	Variable speed
34.	In scientific notation, 0.005 is written as:	5×10^{-3}	5×10^3	5×10^{-2}	5×10^{-4}
35.	A raindrop falling from a cloud reaches a constant speed before	Its acceleration	Its weight becomes zero.	Air resistance	Gravity no longer

	hitting the ground. At this stage, the raindrop has reached terminal velocity because:	n becomes zero.		disappears	acts on it.
36.	A spanner of length 30cm uses a force of 150N to open a tight nut. The torque is:	10 N m	25 N m	35 N m	45 N m
37.	What is a reasonable estimate for the mass of an apple?	2kg	200g	2mg	20kg
38.	A relationship between physical quantities remains unchanged under the same conditions after many experiments. Scientists accept it as a:	Theory	Assumption	Hypothesis	Law
39.	Which of the following is a physical property that can be used to measure temperature?	mass of a liquid	volume of a liquid	mass of a substance	weight of a substance
40.	A perpetual motion machine is impossible because of:	Friction	Conservation of energy	Gravity	Air resistance

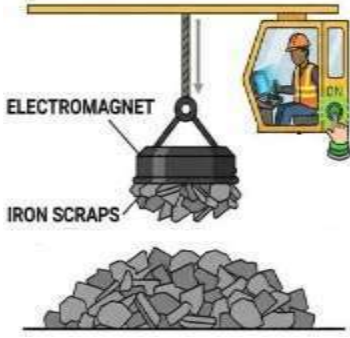
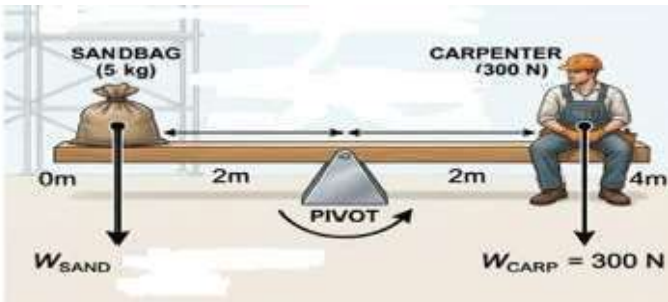
STIMULUS/SITUATION/SCENARIO BASED MCQs

Scenario for MCQs # 41 to 44

Atif is playing with his remote-controlled car on a flat carpet. When he gently pushes the throttle forward, the car starts moving and gradually speeds up on a straight line. Curious, he increases the force applied by the motor to observe how the motion changes.



41.	What type of motion does the toy car exhibit?	constant velocity	constant acceleration	decreasing velocity	decreasing momentum
42.	How would the toy car's speed-time graph look like?	Straight line with negative slope	Straight line with positive slope	curved graph	Horizontal line
43.	The toy car has a mass of 0.5 kg. If it accelerates at 0.4 m/s^2 , what is the magnitude of the force applied to it?	0.2 N	0.4 N	0.8 N	2 N
44.	He then places a small weight on the car, increasing its mass while applying the same force. What will happen to acceleration?	increases	decreases	remains same	becomes zero

	Scenario for MCQs # 45 to 48	A worker in a scrapyard is operating a large crane fitted with an electromagnet. His job is to load and unload heavy iron scrap from trucks. 			
45.	What key idea should the worker keep in mind while operating the crane with an electromagnet?	Electro-magnet works without electricity	Magnet can be controlled (ON/OFF)	Electro-magnet only attracts plastic materials	Magnet becomes weaker as iron is attached to it
46.	Why does the crane attract only iron scraps and not plastic or wood?	Plastic and wood are lighter than iron	Plastic and wood are electrically charged	Iron is a magnetic material	Both plastic and wood block magnetic field
47.	What will happen to the crane's ability to lift and drop scrap if he replaces the electromagnet with a permanent bar magnet?	It will only attract iron pieces	It works as normal	It will attract only plastic materials	It will repel all types of materials
48.	Why is an electromagnet preferred over a permanent magnet in a scrapyard crane?	It is cheaper than a permanent magnet	It allows control over lifting and releasing scrap	It attracts all metals permanently	It does not use any electricity
	Scenario for MCQs # 49 to 52	A carpenter is working on a construction site. He places a 5 kg sandbag on one end (left side) of a 4 m long uniform wooden beam. The beam is supported at a pivot. He then sits on the opposite side (right side) to balance the beam so that it remains horizontal. 			
49.	The situation described is an example of:	Circular motion	Principle of moments	Principle of inertia	Principle of pulley

50.	Which factor does not affect the moment of a force?	Shape of the object	Magnitude of force	Distance from pivot	Direction of force
51.	If the 5 kg sandbag is now placed 1.2 m on the left side of the pivot, how far must he sit on the right side to balance the beam [$g=10\text{ms}^{-2}$]?	0.8 m	1.2 m	2 m	0.2 m
52.	The carpenter shifts the pivot so it is now 1 m from the sandbag (left side). What is the moment arm of carpenter?	1 m	2m	3 m	4 m
Scenario for MCQs #53 to 56		A space research organization is launching a new satellite into space from earth.			
53.	Which expression will be used to determine the orbital velocity of satellite around the earth?	$\frac{2\pi r}{t}$	$\frac{2\pi r}{T}$	$\frac{2\pi T}{R}$	$\frac{2\pi t}{r}$
54.	The period of satellite to orbit the earth is 96.5 minutes, at a height $6.97 \times 10^6\text{m}$ from center of earth. What will be its speed in the orbit?	7.56 km/s	7.56 m/s	756 km/s	7.56 cm/s
55.	The satellite is launched through a rocket. The engine is started, hot gases are pushed downwards and the rocket lifts off the ground. Which force is directly responsible for pushing the rocket upwards?	The friction between rocket and air.	The force of engine downwards.	The force exerted by the exhaust gases pushing against the ground.	The reaction force exerted by the exhaust gases on the rocket.
56.	Which of the following explains the working of a rocket in space?	Pascal's law	Buoyant force	Newton's third law of motion	Thermal expansion of gases
Scenario for MCQs #57 to 60		Amir is helping his father in the garden. After digging, they need to walk across a snowy lawn to reach the tool shed. Amir is wearing his sports shoes while his father is wearing flat snowshoes.			
57.	Amir steps onto the snow and his feet immediately sink deep while his father walks easily on the surface. What is the physical reason behind this?	The snowshoes increase the weight on snow.	The snowshoes decrease the weight on snow	The snowshoes increase the area of contact, decreasing pressure	The snowshoes decrease the area of contact, decreasing pressure
58.	For a constant force, what is the relation between the pressure exerted and area of contact?	direct	exponential	proportional	inverse

59.	Amir's weight is 600 N, and area of contact of his sport shoes with snow is 0.02 m^2 . What is the pressure exerted by Amir on the snow?	12 kPa	30 kPa	600 Pa	30 Pa
60.	Amir's father has the same weight (600 N) and his snowshoes contact area is 0.20 m^2 . How many times smaller is the pressure, he exerts as compared to Amir?	20 times	2 times	5 times	10 times